Machine Learning Experimentation and Model Evaluation Essay

Mubashira Qari

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Essay: Machine Learning Model Experimentation and Evaluation

Introduction

The goal of this project was to develop, evaluate, and compare machine learning models to predict whether a customer would subscribe to a term deposit based on the *Bank Marketing* dataset.
This dataset contains both categorical and numerical features describing customer demographics, past interactions, and macroeconomic indicators.
The experiments aimed to assess model accuracy, generalization, and sensitivity to class imbalance, which is a common issue in marketing datasets where the "yes" class (subscribed customers) is much smaller than the "no" class.

To systematically explore performance differences, three types of algorithms were selected — **Decision Tree**, **Random Forest**, and **AdaBoost** — each trained in both **baseline** and **tuned** configurations.
The experiments were designed to highlight how different learning paradigms handle the bias–variance trade-off, class imbalance, and model stability.
Performance was evaluated using **Accuracy**, **Precision**, **Recall**, **F1-score**, and **AUC (Area Under the ROC Curve)**, providing a comprehensive view of both predictive strength and class discrimination ability.

Experiment Design and Rationale

Why These Experiments Were Chosen

The selection of experiments was guided by three core objectives:
1. **Model diversity:** to compare simple interpretable models (Decision Trees) with advanced ensemble methods (Random Forest, AdaBoost).
2. **Bias–variance exploration:** to study how each model type balances simplicity and generalization.
3. **Impact of tuning:** to assess how hyperparameter adjustments and data balancing affect predictive performance.

The **Decision Tree** was chosen for its interpretability and sensitivity to overfitting, making it ideal for understanding the bias–variance dynamics.
Random Forest was selected as an ensemble of trees that reduces variance by averaging predictions, providing a benchmark for stability and robustness.
Finally, **AdaBoost** was included as a boosting algorithm that sequentially adjusts weights to focus on difficult-to-classify examples — a valuable contrast to bagging methods like Random Forest.

Each model was evaluated twice:
- A **Baseline (A)** version, trained on the original imbalanced dataset with default parameters.

- A **Tuned (B)** version, trained using data preprocessing (such as SMOTE balancing) and parameter optimization (like tuning depth or boosting iterations) to improve minority class detection.

Results Summary

Algorithm	Experiment	What Changed	Accuracy	Precision	Recall	F1	AUC
Decision Tree	DT-1A Baseline	No tuning, imbalanced data	0.8988	0.7130	0.1697	0.2741	0.7082
Decision Tree	DT-1B Tuned	SMOTE-balanced, cp=0.005, maxdepth=6	0.8117	0.8580	0.7187	0.7822	0.8427
Random Forest	RF-2A Baseline	Default parameters, no CV	0.8986	0.6139	0.2674	0.3726	0.7870
Random Forest	RF-2B Tuned	3-fold CV, mtry={3,5}, ntree=150	0.8995	0.6410	0.2451	0.3547	0.7864
AdaBoost	AB-3A Baseline	adabag::boosting, mfinal=30	0.9003	0.6970	0.2035	0.3150	0.8071
AdaBoost	AB-3B Tuned	Factor alignment, tuned mfinal=35	0.8973	0.5997	0.2638	0.3665	0.7761

Bias and Variance Discussion

Decision Tree

The **baseline Decision Tree (DT-1A)** displayed high bias and low variance — it was too simple to capture complex relationships, yielding a poor Recall (0.17).
Although accuracy appeared high, it reflected the model's bias toward predicting the majority "no" class, a common issue in imbalanced data.
After **SMOTE balancing and hyperparameter tuning (DT-1B)**, bias decreased significantly and the model became more flexible. Recall improved to **0.72**, and AUC increased to **0.84**, showing that the tuned Decision Tree effectively captured patterns from both classes.
However, as flexibility increased, variance also rose slightly — the model became more sensitive to fluctuations in training data. Nevertheless, this was an acceptable trade-off for improved minority detection.

Random Forest

Random Forest inherently reduces variance through bagging and random feature selection. The **baseline RF-2A** achieved a strong balance with **Accuracy = 0.8986** and **AUC = 0.7870**, confirming its robustness. Tuning the number of trees and variables (`ntree=150, mtry=3-5`) in **RF-2B** led to only marginal gains in

Precision but no significant Recall improvement.
This indicates that Random Forest already achieves a near-optimal bias–variance balance at default settings. It generalizes well but remains moderately biased toward the majority class.
In practice, this stability makes Random Forest suitable for consistent predictions, though not necessarily the best for minority detection.

AdaBoost

AdaBoost (Adaptive Boosting) focuses on reducing bias by iteratively correcting misclassified samples.
The **baseline AdaBoost (AB-3A)** achieved the **highest overall accuracy (0.9003)** and **AUC (0.8071)** among all baseline models. However, its **Recall (0.20)** was modest, showing limited sensitivity to minority cases.
The **tuned AdaBoost (AB-3B)** fixed factor-level inconsistencies and adjusted iteration parameters for stability. Recall improved to **0.26**, though Precision decreased slightly.
This demonstrates AdaBoost's tendency to reduce bias but increase variance if overtrained. Despite high accuracy, its performance depended heavily on clean, well-preprocessed data.

Optimal Model and Justification

The **tuned Decision Tree (DT-1B)** was identified as the **optimal model** across all experiments.
It achieved the best balance of performance metrics:
- **Highest Recall (0.7187)** – excellent sensitivity to positive (minority) class.
- **Highest F1-score (0.7822)** – strong harmonic balance between Precision and Recall.
- **Highest AUC (0.8427)** – superior ability to distinguish between positive and negative classes.

Although its overall Accuracy (0.8117) was slightly lower than AdaBoost and Random Forest, the improvement in Recall and F1 makes it the most valuable model for the problem context.
In marketing analytics, the cost of missing a potential customer (false negative) is typically greater than contacting a disinterested one (false positive). Thus, Recall and AUC are prioritized over sheer accuracy.

Conclusions and Recommendations

This experimentation revealed several important insights:

1. **Bias–Variance Trade-off:**
    - The baseline Decision Tree suffered from high bias and underfitting.
    - Random Forest successfully reduced variance but introduced moderate bias.
    - AdaBoost minimized bias but increased variance and required careful tuning.
2. **Impact of Data Balancing:**
   The application of **SMOTE** dramatically improved Recall and AUC for the Decision Tree, proving that resampling is crucial for imbalanced datasets.
3. **Model Interpretability vs. Performance:**
   Decision Trees provide interpretability and actionable insights, making them ideal for business applications.
   Ensembles (RF and AdaBoost) offer higher stability but act as "black boxes," which can limit transparency.

Final Recommendation

The **Tuned Decision Tree (DT-1B)** is recommended as the **optimal model**.
It achieved the best trade-off between sensitivity, interpretability, and performance.
For operational deployment, organizations seeking higher robustness could complement it with a **Random Forest** ensemble for validation.
If computational resources permit, a **hybrid ensemble** combining SMOTE balancing and AdaBoost could further improve detection of minority outcomes.

In summary, the experiments demonstrate that **careful tuning, bias–variance management, and proper handling of data imbalance** are essential to developing models that not only perform well statistically but also generalize effectively to real-world marketing scenarios.

Word Count: ~860 words